THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** 0.53	3400/6400.
3	**(b)** flooding by rising sea levels	Higher sea level.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** 42 million cubic km	0.03 x 1.4 billion.
6	**(b)** using public transport and saving electricity	Reduces emissions.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(b)** pressure raises the melting point	High pressure keeps it solid.
9	**(b)** the water table falls	Overdraft.
10	**(c)** A is true but R is false.	A is true; R is false: the Moon has almost no atmosphere.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	0.15 x 1.08 x 10^12 (1); = 1.6 x 10^11 cubic km (1).	1+1
12	Air is extremely thin (1); Gases gradually escape to space (1).	1+1
13	Air is thinner (1); Less heat is held, and air cools with height (1).	1+1
14	It radiates back as much as it receives (1); The atmosphere regulates the balance (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dense iron sank to the centre (1); Lighter material floated up (1); Seismic S-waves do not pass through the outer core (1).	1+1+1
16	Early atmosphere had CO2 and water vapour (1); Photosynthesis added oxygen (1); Ozone layer formed and life spread on land (1).	1+1+1
17	Day length would be about 12 hours (1); Shorter days and nights (1); Winds and weather would change (1).	1+1+1
	OR: 70/6400 x 100 (1); = 1.1 per cent (1); Crust is a very thin skin (1).	1+1+1
18	Lithosphere: soil and minerals (1); Hydrosphere: water (1); Atmosphere: oxygen and CO2 (1).	1+1+1
	OR: Weathering breaks rocks (1); Organic matter mixes (1); Plants grow and support food chains (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Inner core = 6400 - (35 + 2900 + 2200) (1); = 1265 km (1); Crust 0.5 per cent, mantle 45 per cent (1); Outer core 34 per cent, inner core 20 per cent (1); Mantle is the thickest (1).	1+1+1+1+1
	OR: Atmosphere: warming (1); Oceans: more acidic, coral damage (1); Plants: faster growth up to a point (1); But heat and drought reduce crops (1); Net effect is harmful (1).	1+1+1+1+1
20	Venus: strong greenhouse and closer to Sun (1); Earth: moderate greenhouse (1); Mars: thin atmosphere, farther from Sun (1); Atmosphere composition matters (1); Balance is delicate for habitability (1).	1+1+1+1+1
	OR: Ozone measured by instruments (1); Forest cover through images (1); Glaciers retreat mapped (1); Cyclones tracked (1); Benefits: warnings, planning (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Venus	Runaway greenhouse.
2	**(b)** about one-third the oxygen	Pressure proportional to density.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(d)** increase greatly	Like the Moon.
5	**(c)** there are very few particles to transfer heat	Very low density.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** seismograph	Records vibrations.
8	**(a)** burning fossil fuels	Industries and vehicles.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(c)** carbon dioxide	Combustion.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Pressure inside our body balances it (1); It acts equally from all directions (1).	1+1
12	Cold polar clouds help CFC reactions (1); Sunlight returns and triggers ozone destruction (1).	1+1
13	Natural effect keeps the Earth about 33 degrees Celsius warmer (1); Extra gases trap more heat, causing global warming (1).	1+1
14	Area = 10 x 10^6 square m (1); Volume = 600/1000 x 10^7 = 6 x 10^6 cubic m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	More heat trapped (1); Ice melts, sea level rises (1); Extreme weather and crop losses (1).	1+1+1
16	Water cycle renews it (1); Fresh water is little and unevenly distributed (1); Pollution makes it unusable (1).	1+1+1
17	Smog and respiratory diseases in cities (1); Stratosphere: ozone depletion (1); Different layers, different problems (1).	1+1+1
	OR: Water has a high heat capacity (1); Coasts have mild climates (1); Oceans dissolve CO2 and marine plants use it (1).	1+1+1
18	Forests: photosynthesis (1); Wetlands: store carbon in soil (1); Oceans: dissolve CO2 (1).	1+1+1
	OR: Store fresh water as ice (1); Feed rivers in summer (1); Millions depend on them (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lithosphere: new rock and soil (1); Atmosphere: ash and gases, temporary cooling (1); Hydrosphere: acid rain, changed water (1); Biosphere: destruction and then fertile soil (1); Recovery over years (1).	1+1+1+1+1
	OR: 20/6400 x 100 (1); = about 0.3 per cent (1); Life is a thin layer (1); Depends on delicate conditions (1); Protecting it is important (1).	1+1+1+1+1
20	List electricity, transport, waste, paper use (1); Collect monthly data (1); Convert to approximate CO2 (1); Identify the largest sources (1); Actions: solar panels, carpool, recycle (1).	1+1+1+1+1
	OR: Falling groundwater (1); Johads store rain (1); Baoris (stepwells) conserve water (1); Drip irrigation saves water (1); Reuse of wastewater and awareness (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** without CO2 the Earth would be too cold and plants could not make food	Natural greenhouse effect.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** decreasing, increasing, decreasing	Troposphere, stratosphere, mesosphere.
4	**(b)** large energy is needed to remove salt	Energy intensive.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(b)** its axis is tilted similarly	Tilt about 25 degrees.
7	**(b)** Earth's magnetic field	Magnetic sense.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** freezing of most water	Too cold.
10	**(d)** from deep oceans to the lower atmosphere	Life zone.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Compare atmospheres and climates (1); Understand what makes Earth habitable (1).	1+1
12	Ice melts, exposing darker surfaces (1); Dark surfaces absorb more heat (1).	1+1
13	Land ice adds new water to the sea (1); Floating ice already displaces its own weight of water (1).	1+1
14	Total greenhouse gases we cause (1); Reducing it limits global warming (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Stratospheric ozone absorbs UV, is beneficial (1); Ground-level ozone is formed from pollutants (1); It harms lungs and plants (1).	1+1+1
16	Exploration yields knowledge and technology (1); May provide new resources (1); Earth's problems need urgent care (1).	1+1+1
17	No seasons (1); Same day length (1); Poles colder, tropics hotter all year (1).	1+1+1
	OR: Pressure decreases rapidly (1); Temperature decreases about 6.5 degrees Celsius per km (1); Less air above at height (1).	1+1+1
18	Low-lying coasts flood (1); Sea walls, mangroves (1); Early warning and relocation (1).	1+1+1
	OR: Only known planet with life (1); Resources are limited (1); Future generations depend on it (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Burning fossil fuels raised CO2 (1); CFCs depleted ozone (1); Global warming and sea rise (1); Acid rain and smog (1); Societies: health, farming, migration (1).	1+1+1+1+1
	OR: Atmosphere: CO2 has risen due to fuels and the ozone layer was damaged by CFCs (1); Hydrosphere: rivers and seas are polluted and glaciers are melting (1); Biosphere: forests are cleared and species are lost (1); The biosphere is a thin layer and sensitive (1); Hence the claim is wrong; action is needed (1).	1+1+1+1+1
20	Moderate temperature allows liquid water (1); Closer: more evaporation, greenhouse runaway (1); Farther: oceans freeze (1); Atmosphere regulates temperature (1); The habitable zone is narrow (1).	1+1+1+1+1
	OR: Plants take CO2 from air (1); Animals and decomposers return CO2 (1); Oceans store carbon (1); Fossil fuels release stored carbon (1); Excess CO2 disturbs climate and oceans (1).	1+1+1+1+1

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